FOROH — Formulation, Positioning & Design Notes

Scope. Part I gives the formal definitions. Part II gives the design intent, prior-art positioning, and open validation tasks. Claims are separated into *established* and *assumption*; assumptions are stated to be tested, not closed.

Part I — Formulation

1. Coordinates

$$z \in \mathbb{R}^{d_b} \xrightarrow{g_\phi} \tilde{u} \in \mathbb{R}^d, \qquad u = \frac{\tilde{u}}{\|\tilde{u}\|} \in S^{d-1} \tag{1}$$

$$w = \frac{w_{\text{raw}}}{\|w_{\text{raw}}\|} \in S^{d-1}, \qquad w_{\text{raw}} \in \mathbb{R}^d \text{ (learnable)} \tag{2}$$

Spherical decomposition (polar θ, azimuthal v):

$$u = \cos\theta\, w + \sin\theta\, v, \qquad \theta = \arccos(u \cdot w) \in [0, \pi], \qquad v = \frac{u_\perp}{\|u_\perp\|} \in S^{d-2},\ v \perp w \tag{3}$$

$$u_\perp = u - (u \cdot w)\, w \tag{4}$$

Learnable parameters: ϕ, w_{raw} only. θ carries the label component; v carries the rest.

2. Static ordinal (reads θ only)

$$\hat{y} = \frac{\theta}{\pi} C_{\max}, \qquad \hat{c} = \text{round}(\hat{y}) \tag{5}$$

Target ring of grade y:

$$\theta_y = \frac{\pi y}{C_{\max}}, \qquad \mathcal{R}_y = \{\cos\theta_y\, w + \sin\theta_y\, v' : v' \in S^{d-2}\} \tag{6}$$

Proposition 1.

$$d_g(u, \mathcal{R}_y) = \min_{t \in \mathcal{R}_y} \arccos(u \cdot t) = |\theta - \theta_y| = \frac{\pi}{C_{\max}} |\hat{y} - y| \tag{7}$$

Proof. $u \cdot t = \cos\theta \cos\theta_y + \sin\theta \sin\theta_y\, (v \cdot v')$. With $\theta, \theta_y \in [0, \pi]$, the factors $\sin\theta, \sin\theta_y \geq 0$, so $u \cdot t$ is maximized at $v \cdot v' = 1$, giving $u \cdot t = \cos(\theta - \theta_y)$. Since arccos is decreasing, the minimum distance is $\arccos\cos(\theta - \theta_y) = |\theta - \theta_y|$. □

Gradient acts only along the meridian ($\partial/\partial v = 0$).

3. Loss (Huber)

Let $e = \hat{y} - y$:

$$L_\delta(e) = \begin{cases} \dfrac{e^2}{2\delta} & |e| \le \delta \\ |e| - \dfrac{\delta}{2} & |e| > \delta \end{cases} \qquad L'_\delta(e) = \begin{cases} \dfrac{e}{\delta} & |e| \le \delta \\ \mathrm{sign}(e) & |e| > \delta \end{cases} \tag{8}$$

4. arccos gradient (implicit curriculum)

Let $s = \cos\theta = u \cdot w$:

$$\frac{\partial \hat{y}}{\partial s} = \frac{C_{\max}}{\pi}\frac{d\theta}{ds} = -\frac{C_{\max}}{\pi}\frac{1}{\sqrt{1-s^2}} = -\frac{C_{\max}}{\pi \sin\theta} \tag{9}$$

$$\frac{\partial L}{\partial s} = L'_\delta(e)\,\frac{\partial \hat{y}}{\partial s} = -\,L'_\delta(e)\,\frac{C_{\max}}{\pi \sin\theta} \tag{10}$$

Magnitude $\propto 1/\sin\theta$: amplified at extreme grades ($\theta \to 0, \pi$), minimal at mid grade ($\theta = \pi/2$). The intrinsic spherical gradient is position-independent; the $1/\sin\theta$ factor arises from the curvature of the $\tilde{u}$ coordinate that backprop traverses.

5. Geometric uncertainty (from θ)

$$\|u_\perp\| = \sqrt{1-(u\cdot w)^2} = \sin\theta, \qquad (u\cdot w)^2 + \|u_\perp\|^2 = 1 \tag{11}$$

With orthogonal energy ρ fixed, reachable score range:

$$\hat{y} \in \frac{C_{\max}}{\pi}\left[\arccos\sqrt{1-\rho^2},\ \pi - \arccos\sqrt{1-\rho^2}\right] \tag{12}$$

$\rho = 0$: full range (confident). $\rho = 1$: fixed at center $C_{\max}/2$ (fully uncertain). Range is symmetric about $C_{\max}/2$.

6. Score-system agnostic

Same trained θ; swap C at inference:

$$\hat{y}^{(C)} = \frac{\theta}{\pi}C \quad \Longrightarrow \quad \hat{y}^{(3)},\ \hat{y}^{(4)},\ \hat{y}^{(100)} = \frac{\theta}{\pi}\cdot\{3, 4, 100\} \tag{13}$$

7. Temporal (reads v; no loss signal on v)

A sequence traces a curve on S^{d-1}:

$$u(t) = \cos\theta(t)\,w + \sin\theta(t)\,v(t), \qquad v(t) \in S^{d-2} \tag{14}$$

$\theta(t)$: severity over time. $v(t)$: path component distinguishing equal-severity sequences. A scalar score retains only $\theta(t)$, so $0\to1\to0$ and $0\to1\to2\to0$ collapse; $v(t)$ separates them on S^{d-2}.

The observed grade round($\hat{y}$) carries the wobble (it is read off θ); v is the reliability cue used to decide which θ jumps are genuine. A magnitude cue fails because $\|u_\perp\| = \sin\theta$ is also the uncertainty (§5): at the spine ($\theta \approx \pi/2$) the cue and the uncertainty coincide.

Assumption (stated, to test). $\hat{\theta}$ and $\hat{v}$ share temporal regularity in degree, not in content. With the $1/\sin\theta$ amplification (§4) inflating $\hat{\theta}'$ at the spine but not $\hat{v}'$, the match is conditioned on θ and the amplification removed in closed form by the known factor $\sin\theta$:

$$P\big(\hat{v}'(t)\,\big|\,\theta\big) \;\approx\; P\Big(\sin\theta\,\hat{\theta}'(t)\,\Big|\,\theta\Big) \tag{15}$$

θ is the only conditioning variable: it is the existing coordinate, the amplification is a function of θ alone, and $\rho = \sin\theta$ is the same quantity. No parameter, no new axis. A timepoint with large $\hat{\theta}'$ but small $\sin\theta\,\hat{\theta}'$ is amplification-driven (noise); large after normalization is genuine.

Why $\approx$ and not $=$ (clinical rationale). The match is approximate *by assumption*, not by finite-sample error — enlarging the sample does not drive it to equality. θ and v are read from the *same* timepoint by the same encoder, so they should change with shared temporal regularity: a genuine change moves both smoothly, noise perturbs both. They share the *degree* of regularity while their *content* differs (θ = amount of progression, v = path direction). This single statement in fact bundles three premises, which the spine test (§8) cannot fully separate:

1. **Sharing:** same-source coordinates share temporal regularity.
2. **Separation:** they share *degree* but differ in *content* — otherwise v is a copy of θ and carries no new information.
3. **Noise symmetry:** noise enters θ and v through comparable channels. If noise is asymmetric (e.g. label wobble that perturbs θ only, not v), sharing fails even when (1)–(2) hold. Same-source does not guarantee same noise channel.

Premises (1) and (2) pull against each other (strong sharing weakens the information in v; strong separation weakens the regularity match), so the mechanism operates in a narrow band whose width is data-determined.

8. From cue to regularizer (and its open status)

$$\frac{\partial L_{\text{label}}}{\partial v} = 0 \tag{16}$$

$v(t)$ receives *no label signal.* It is, however, intended to be shaped by a *regularization* term that enforces the §7 match (drive $P(\hat{v}' \mid \theta)$ toward $P(\sin\theta\,\hat{\theta}' \mid \theta)$), applied at fixed strength $\lambda = 1$ (a design choice, not a tuned parameter, so the "no-parameter" property is preserved; ablate by on/off). Note the distinction between *no label loss* and *no regularization*: under this term v is not undetermined but is pushed toward sharing regularity with θ.

This raises the stakes of the §7 assumption. As a passive cue, a wrong assumption merely yields inaccurate judgments. As an active regularizer, a wrong assumption is *written into* the encoder: if the data do not in fact share regularity, the term forces them to match and *distorts* the representation — potentially contaminating the θ (severity) readout, not just the v residual. Active use is stronger but its downside is larger than passive use.

Costs of the assumption: (i) the distributional match needs enough samples per θ band, trading against small-n — and the regularizer makes this sharper, since the strength FOROH relies on (small-n robustness) competes for the same samples the §7 term consumes; (ii) conditioning on θ alone cannot catch wobble that depends on the direction of v — the same price as the single-axis assumption. Whether $\sin\theta\,\hat{\theta}'$ tracks $\hat{v}'$ in distribution near the spine is the test that closes or breaks the assumption; if it breaks, the spine test alone may not distinguish *which* premise of §7 failed (sharing vs. noise symmetry).

Part II — Positioning & Design Notes

9. Core idea (one sentence)

Place embeddings on S^{d-1}, read the monotone progression quantity directly as the polar angle θ off a single axis w, and keep the orthogonal (azimuthal) component v as a free residual. Severity is *defined by geometry*, not learned as parameters.

10. Why "no parameters" is the load-bearing strength

The severity coordinate system is fixed by geometry rather than learned: θ, the rings θ_y, and the score $(\theta/\pi)C$ are all read directly (Part I §1–2); only ϕ and the single raw axis w_{raw} are trained. This single property propagates:

1. **Small-n robustness.** The coordinate system cannot collapse under scarce data because it is not learned. Aligns with the small-cohort setting of interest.
2. **Score-system agnostic.** Because θ is geometric, the grade count C can be swapped at inference without retraining (§6).
3. **Free orthogonal axis.** v is *defined* as the orthogonal component of u, so it carries no *label* loss ($\partial L_{\text{label}}/\partial v = 0$); it is shaped only by the §7–§8 regularizer and used as a reliability tool (§7), not as a separately-learned axis.
4. **Domain-agnostic.** Few modality-specific learned parts $\Rightarrow$ the structure transfers across time-series domains unchanged.

11. Geodesic-to-ring result (established)

The geodesic distance from u to the target ring equals $|\theta - \theta_y|$ (Part I, Proposition 1; maximized when the azimuthal parts align, $v \cdot v' = 1$). Consequence: ordinal distance is *linearized* as an angle difference. Unlike DORGA — where multiple anchors on a great arc are tied to the severity axis by a *soft* ordinal-separation loss, leaving mid-grade collinearity only approximately enforced — here severity is read off a single w as θ, so no inter-anchor constraint exists to be slack: the linearization holds *by construction*, not by optimization.

12. The actual contribution: orthogonal axis as a tool, not a readout

The orthogonal axis v is used as a *reliability cue* to straighten the θ trajectory — to decide which round-boundary crossings are genuine worsening and which are noise. The wobble lives in θ (the observed round($\hat{y}$) oscillates, e.g. $1 \to 3 \to 2 \to 3$, where the monotone reading $1 \to 2 \to 2 \to 3$ is clinically correct); v is not where the wobble sits but the side information that judges it. A magnitude cue collapses at the spine because $\|u_\perp\| = \sin\theta$ is also the uncertainty, so the cue is moved onto temporal regularity, conditioned on θ, with the amplification removed in closed form (Part I §7, eq. for $P(\hat{v}' \mid \theta)$). Honest costs are the small-n trade-off and the v-direction blindness (Part I §8).

13. Positioning vs. Ouyang et al. (TMI 2022)

Ouyang et al. disentangle aging vs. disease as two orthogonal directions in a latent space. This is the closest prior art and **must be cited**. It differs on three axes:

	Ouyang (TMI 2022)	FOROH
Stage	Euclidean latent, two learned vector directions	S^{d-1}, polar angle θ (geometric)
Orthogonal axis	*extracted as the readout* (it *is* severity)	*used as a tool* to stabilize θ
Separation engine	binary label (normal/diseased); needs a control cohort	geometry only; no labels, no control cohort
Generality	medical longitudinal (control-cohort assumptions)	domain-agnostic by construction

Takeaway: not a blocker but a stepping stone. The contribution lives precisely where Ouyang does *not* go: (i) orthogonal axis as a stabilizer, (ii) geometry-only separation requiring neither labels nor a control cohort.

Second prior-art line: consistency-based noise detection. The upper-level idea behind §7–§8 — using the *consistency* of two signals to tell genuine change from noise — is well established in noisy-label learning (two-classifier-head agreement, transform-consistency, co-teaching; e.g. Centrality & Consistency, ECCV 2022; Consistency Regularization for Label Noise, 2021). That literature even treats symmetric vs. asymmetric noise explicitly, matching premise (3) of §7. So the *principle* is not novel and must be cited. What is not found there is FOROH's specific form: consistency between the two *orthogonal components* of a single representation (not two equal predictions), along the *temporal* axis, after a *closed-form* removal of a known geometric amplification ($\sin\theta$). The mechanism is best presented as a component of the geometric framework, not as a standalone noise-detection contribution.

14. Intended direction: a general time-series model

FOROH is meant to work beyond medicine — any progression where compressing to a single monotone quantity destroys information. θ = monotone progression; v = component separating equal-progression sequences. The abstraction is deliberate (the formulation avoids hard-coding "disease").

15. Open tasks / honest limits

1. **Generality is a proof, not a claim.** One domain is not evidence of universality. Need ≥ 2 structurally different time-series domains showing the θ/v split works (cf. CXR + EEG cross-domain replication in the MedIA work).
2. **Sell structure, not peak accuracy.** A general model may lose to a domain-specific one on a single domain. The selling point is the structural properties (no parameters / no assumptions / domain-agnostic), not benchmark wins.
3. **Equal-spacing assumption.** Rings fixed at $\theta_y = \pi y / C_{\max}$ assume roughly uniform ordinal spacing. If real grade gaps are non-uniform, the fixed geometry cannot express it (the price of no parameters). Whether this hurts is data-dependent.

4. **Single-axis (1D) severity assumption.** A single w assumes progression is 1-dimensional. Multi-faceted worsening would not fit a single θ. Stronger assumption than a learned axis.

5. **Temporal-regularity assumption (§7).** The $P(\hat{v}' \mid \theta) \approx P(\sin\theta\,\hat{\theta}' \mid \theta)$ match is stated, not closed; the spine-region test decides it.

6. **Second validation domain: TBD.** Choosing it is the first step that turns "general" from claim into evidence.

7. **Frame the §7–§8 mechanism as a component, not a headline.** As a standalone "self-consistency noise detector" it is dominated by the noisy-label literature (§13). Its value is that the geometry yields it *for free* inside the parameter-free framework; presented that way it strengthens the whole, presented alone it is a special case of known work.

8. **Design direction (planning-stage, form undecided): multi-ROI as a moving surface.** With a single ROI a sequence is a *curve* $u(t)$ on S^{d-1}. With R ROIs (e.g. six lung regions) each timepoint is a set $\{u_r(t)\}$, and over time the bundle sweeps a 2-D *sheet* (time × ROI). The intended reading of that sheet, in terms of how the per-ROI θ_r move:

   - *All ROIs same direction:* the sheet *stretches or shrinks along w* — global progression or recovery. Not a fold. Kept.
   - *One ROI moves, others fixed:* the sheet *tilts* smoothly to one side — *localized* disease (e.g. one lobe worsening). **Not a fold.** Kept; this is exactly the clinically important localized signal, and the control must not touch it.
   - *Sideways spread* (azimuthal, v direction) with little w-progress — the same as high $\rho = \sin\theta$ uncertainty (§5).
   - *Fold (non-physical):* two ROIs move in *opposite θ* directions at the same time — one forward (worsening), another backward (improving), a *sign split* across ROIs. This, and only this, folds the sheet.

   The fold is the sole target: not size, not tilt, not a single ROI moving. A *weak* control penalizing only this opposite-direction split would suppress the non-physical case while leaving global progression, localized worsening, and asymmetric (but same-sign) change all intact. The clinical prior here is narrow and *checked, not assumed*: a senior radiologist (20 years) confirmed that simultaneous opposite movement across lung regions is closer to noise than to a real pattern, and it appears only rarely in the data. So the penalty targets a genuinely non-physical, empirically scarce event — near-zero most of the time, active only at those rare moments. It connects to DORGA: DORGA's belief propagation coupled ROIs by *static grade relations*; this couples them by *dynamic sign coherence* across ROIs. Form (which curvature, measured how) is left open — a direction seen at FOROH's planning stage, not a specified mechanism. The only residual question is whether *any* genuine simultaneous opposite movement exists (e.g. one lobe worsening while another responds to treatment); if so it is rare, and a *weak* penalty damps it without erasing it.

Standing caveats

All strong evaluations rest on two premises carried from prior discussion: that the work is substantively author-driven, and that the core conceptual moves (geometry-only separation, orthogonal-axis-as-tool) are the author's own. Within those premises the positioning above holds.